INCH-POUND
MIL-DTL-24643/40G
27 November 2023
SUPERSEDING
MIL-DTL-24643/40F
1 October 2009

# DETAIL SPECIFICATION SHEET

## CABLE, ELECTRICAL, -4 DEGREES FAHRENHEIT (-20 DEGREES CELSIUS) TO +194 DEGREES FAHRENHEIT (+90 DEGREES CELSIUS), 600 VOLTS, TYPE LS1SMU

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

Construction (Non-Watertight)

| | | |
|---|---|---|
| First | - | Copper conductor, tin-coated, ASTM B286, 22 AWG with 7 strands. |
| Second | - | Separator may be used at manufacturer's option where required to provide free-stripping insulation. |
| Third | - | Thermoset insulation, white or natural. Nominal wall thickness of 0.030 inch; wall thickness may vary from nominal as necessary, in order for completed cable to meet the specified electrical requirements. |
| Fourth | - | Braided shield of 34 or 36 AWG, tin-coated copper. Braid angle of 20 to 25 degrees; minimum coverage of 90 percent. |
| Fifth | - | Shield insulation of two polyester tapes, outer tape sealed. An alternate shield insulation may consist of one polyester tape plus an extruded jacket of a suitable transparent material; minimum average thickness of the transparent material 0.003 inch. Standard identification code shall be applied by method 2. |
| Sixth | - | Five singles cabled around a center filler with a lay in accordance with 3.4.4 of MIL-DTL-24643. Cabling sequence shall be consecutive, starting with pair no. 1, from the center outward. Fillers may be used as necessary to form a firm, well-rounded assembly. |
| Seventh | - | An optional binder. |
| Eighth | - | Cross-linked polyolefin jacket. Absolute minimum average wall thickness 0.063 inch. See table I for cable overall diameter. |

NOTE: Manufacturer's identification tape is required (see 3.4.9 of MIL-DTL-24643).



Distribution Statement A. Approved for public release: distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:10Z
Check the source to verify that this is the current version before use.

TABLE I. Details.

| Military part no. M24643/40 | Type and size | Overall diameter (inch) | |
|---|---|---|---|
| | | min. | max. |
| -01UN | LS1SMU-5 | 0.465 | 0.500 |

REQUIREMENTS:

Qualification required.

INSPECTION:

**Basic electricals:**

| | |
|---|---|
| Conductor resistance (ohms/1,000 feet at 77 °F [25 °C], max.) | 17.37 |
| Voltage withstand (volts, root mean square, min.) | |
| Conductor to shield | 5,000 |
| Component shield to component shield | 500 |
| Insulation resistance (megohms/1,000 feet, min.) | |
| Conductor to shield | 500 |
| Component shield to component shield | 100 |
| Conductor continuity | No failure |
| Shield continuity | No failure |

**Group A:**

| | |
|---|---|
| Visual and dimensional | No failure |
| Characteristic impedance, at 1 megahertz (MHz) (ohms) | 50±2 |

**Group B:**

| | |
|---|---|
| Thermoset proof test (percent, max.) | |
| Insulation | 50 |
| Jacket (when tested at 392 °F [200 °C]) | 50 |
| Attenuation, at 400 MHz (decibel [dB]/100 ft max.) | 18 |
| Physicals (unaged) | |
| Insulation (extruded insulation only) | |
| Tensile strength ($lb/in^2$, min.) | 700 |
| Elongation (percent, min.) | 150 |



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:10Z
Check the source to verify that this is the current version before use.

| | |
|---|---|
| Jacket (cable) | |
| Tensile strength (lb/in$^2$, min.) | 1,300 |
| Elongation (percent, min.) | 160 |
| Tear (lb/in thickness, min.) | 35 |
| Cold bending (cable) | No failure |

**Group C:**

| | |
|---|---|
| Physicals (aged) (air oven) | |
| Insulation (extruded insulation only) | |
| Tensile strength (percent of unaged, min.) | 75 |
| Elongation (percent of unaged, min.) | 75 |
| Jacket (cable) | |
| Tensile strength (percent of unaged, min.) | 60 |
| Elongation (percent of unaged, min.) | 60 |
| Permanence of printing (jacket) (cycles, min.) | 125 |
| Heat distortion (percent of unaged, max.) | 30 |
| Shrinkage | No failure |
| Shield (conformance to material, construction, and coverage) | No failure |

**Group D:**

| | |
|---|---|
| Flame propagation (cable) | No failure |

**Qualification inspection:**

Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:

| | |
|---|---|
| Aging and compatibility (cable) (257±9 °F [125±5 °C]) | No failure |
| Abrasion resistance (jacket) (scrapes, min.) | 75 |
| Acid gas equivalent (percent, max.) | |
| Jacket | 2 |
| Fillers | 2 |
| Insulation | 18 |
| Shield insulation | 2 |
| Halogen content (percent, max.) | |
| Jacket | 0.2 |
| Fillers | 0.2 |
| Insulation | 0.2 |



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:10Z
Check the source to verify that this is the current version before use.

| | |
|---|---|
| Immersion (jacket) | |
| Tensile strength (percent of unaged, min.) | 50 |
| Elongation (percent of unaged, min.) | 50 |
| Smoke index (max.) | |
| Jacket | 25 |
| Fillers | 45 |
| Insulation | 45 |
| Toxicity index (max.) | |
| Jacket | 5 |
| Fillers | 5 |
| Insulation | 1.5 |
| Durometer (jacket) – type A (hardness, min.) | 80 |
| Weathering (jacket) | No failure |
| Electrical moisture absorption | No failure |

UNIT ORDERING LENGTHS:

| Type and size | Feet (nominal) |
|---|---|
| LS1SMU-5 | 1,000 |

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

## CONCLUDING MATERIAL

Custodians:
Army – MI
Navy – SH

Preparing activity:
Navy – SH
(Project 6145-2013-056)

Review activities:
Army – AR, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:10Z
Check the source to verify that this is the current version before use.